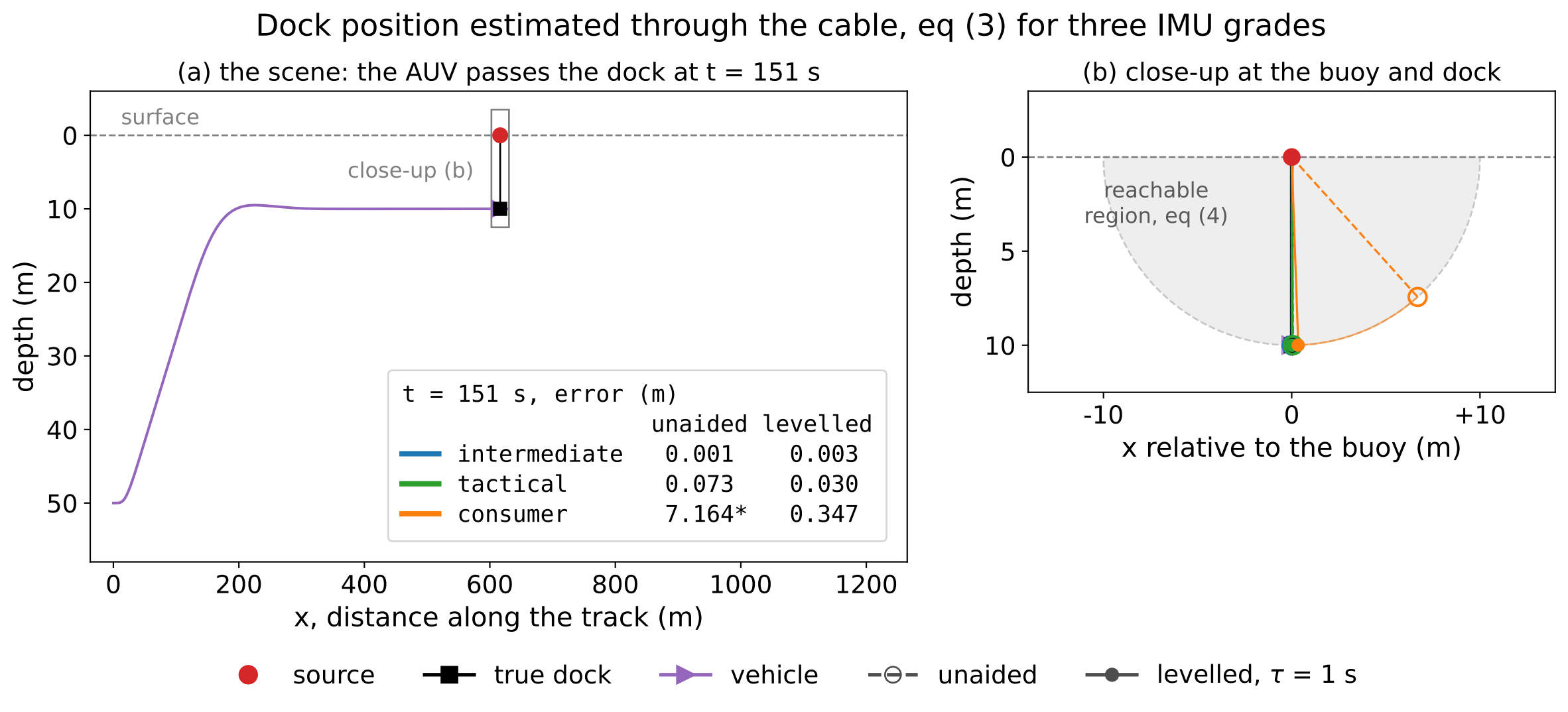
Dock position estimated through the cable, eq (3) for three IMU grades
(a) the scene: the AUV passes the dock at t = 151 s
surface
close-up (b)
depth (m)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
x, distance along the track (m)
t = 151 s, error (m)
unaided levelled
intermediate 0.001 0.003
tactical 0.073 0.030
consumer 7.164* 0.347
(b) close-up at the buoy and dock
reachable
region, eq (4)
depth (m)
0
5
10
-10
0
+10
x relative to the buoy (m)
source
true dock
vehicle
unaided
levelled, τ = 1 s